

# SOWJANYA PRIYA BANDARU

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## SUMMARY

- Data Analyst with over 3 years of experience in data analysis, predictive modelling, and data visualization.
- Skilled in Python, SQL for data processing, cleaning, and developing machine learning models to support business decisions.
- Expertise in statistical analysis, A/B testing, time series analysis, and machine learning techniques such as Logistic Regression, Random Forest, and Naive Bayes.
- Experienced in creating interactive dashboards with Power BI and Tableau, delivering real-time business insights and visual analytics.
- Strong background in healthcare analytics, e-commerce, and customer service, optimizing processes and improving customer engagement.
- Proficient in AWS cloud platforms (Redshift, S3) and data engineering tools like Apache Airflow for seamless data integration and workflow automation.
- Well-versed in Natural Language Processing (NLP) for sentiment analysis and improving chatbot performance based on customer feedback.
- Demonstrated ability to build predictive models for fraud detection and patient care optimization, driving risk reduction and better outcomes.
- Collaborative and effective team player, working with engineers, healthcare providers, and marketing teams to align insights with strategic goals.

## SKILLS

|                                 |                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages           | Python, SQL                                                                                                                                        |
| Databases & Data Warehouses     | PostgreSQL, SQL Server, MySQL, Snowflake, Teradata, BigQuery                                                                                       |
| Data Visualization              | Power BI, Tableau, Advanced Excel                                                                                                                  |
| Libraries & Frameworks          | NumPy, Pandas, Matplotlib, Plotly, Seaborn, Streamlit, FastAPI, NLTK                                                                               |
| Data Engineering                | Alteryx, Talend, Informatica, Apache Airflow, dbt                                                                                                  |
| Cloud Platforms                 | AWS (Redshift, DynamoDB, S3, Athena, Lambda)                                                                                                       |
| Methodologies                   | SDLC, Agile (Scrum), Waterfall                                                                                                                     |
| Version Control                 | Git, GitHub                                                                                                                                        |
| ML Techniques                   | Regression, KNN, Decision Tree, Naive Bayes, Random Forest, LLM                                                                                    |
| Healthcare Tools & Technologies | HIPAA, EMR, HER                                                                                                                                    |
| Analytical Skills               | Data Mining, Data Cleansing, Statistical Analysis, Data Wrangling, Time Series Analysis, A/B Testing, Text Mining, ETL, Data Warehousing, SAP, SAS |

## WORK EXPERIENCE

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| CVS Health<br>Data Analyst                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Feb 2025 – Present  |
| <ul style="list-style-type: none"><li>• Developed machine learning models to detect fraudulent activities by analysing historical claims data and transactional patterns.</li><li>• Pre-processed and cleaned large datasets using Python and SQL, ensuring high-quality data for training AI models and accurate fraud detection.</li><li>• Worked with cross-functional teams to integrate fraud detection models into the claims processing system, reducing false positives and improving detection efficiency.</li></ul>                                                                                                                                                                                                               |                     |
| CVS Health<br>Data Analyst – Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jul 2024 – Dec 2024 |
| <ul style="list-style-type: none"><li>• Created predictive models to identify at-risk patients and optimize care strategies, leveraging historical health data and patient demographics.</li><li>• Cleaned, pre-processed, and analysed large datasets from various healthcare systems with Python and SQL, ensuring high-quality data for model training and analysis.</li><li>• Utilized machine learning techniques such as Logistic Regression and Random Forest to build models that predict patient outcomes, enabling proactive intervention.</li><li>• Collaborated with healthcare providers and stakeholders to integrate predictive insights into patient care workflows, improving care quality and patient outcomes.</li></ul> |                     |
| Dell Technologies<br>Data Analyst                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Jan 2021 – Dec 2022 |
| <ul style="list-style-type: none"><li>• Led the analysis of customer service interactions and designed data pipelines to enhance chatbot performance and ensure more accurate responses.</li><li>• Leveraged Python and SQL to clean and pre-process large datasets from customer service logs, enabling the chatbot to respond to user queries with higher efficiency.</li><li>• Created and implemented NLP (Natural Language Processing) models to analyse customer queries and improve the chatbot’s understanding of context, resulting in a more intuitive user experience.</li></ul>                                                                                                                                                 |                     |

- Incorporated machine learning algorithms to continually optimize the chatbot’s performance and accuracy by learning from real-time user interactions.
- Worked with cross-functional teams, including engineers and customer support, to ensure seamless integration of the chatbot into the existing customer service system.
- Evaluated customer feedback data to identify common issues and improve the chatbot's responses, enhancing overall customer satisfaction by 20%.
- Designed and maintained Power BI dashboards to track chatbot interactions, providing insights on user engagement, issue resolution time, and chatbot effectiveness.
- Conducted A/B testing to compare different versions of the chatbot, using statistical analysis to identify the most effective changes for better user interactions.
- Collaborated closely with business stakeholders to ensure the chatbot aligned with company goals, improving operational efficiency and reducing the workload on customer support agents.

TatvaSoft

Data Analyst

Mar 2020 – Dec 2020

- Analysed customer purchase patterns, browsing behaviour, and demographic data to create actionable insights for personalized marketing strategies.
- Leveraged Python, SQL, and EHR data to collect, clean, and transform large datasets from multiple e-commerce platforms, ensuring high-quality data for analysis.
- Designed and maintained interactive Tableau dashboards, providing real-time insights into customer trends and engagement, influencing key marketing decisions.
- Conducted detailed segmentation analysis using clustering techniques and customer cohort analysis to better target campaigns and improve customer retention.
- Implemented statistical techniques such as A/B testing to evaluate the effectiveness of various marketing strategies, resulting in a 15% improvement in conversion rates.
- Collaborated with cross-functional teams to integrate findings into the company's marketing and customer engagement workflows, enhancing personalization efforts.

THESIS

Efficient Federated Learning Algorithms for Resource - Constrained IoT Devices

Jan 2024 - Dec 2024

- Conducted research on federated learning algorithms for IoT devices, focusing on energy efficiency, communication optimization, and privacy, while evaluating machine learning models.
- Developed scalable federated learning strategies for real-world applications like smart cities, analysing trade-offs between model complexity, accuracy, and resource utilization.

PROJECTS

Data Cleansing, Sales Analysis, and Dashboard Development (SQL, Power BI, Tableau)

- Developed a portfolio project showcasing SQL-based data cleansing, transformation, and sales analysis, ensuring data accuracy and consistency through advanced SQL functions
- Integrated an Excel sales budget dataset into Power BI for a dynamic Sales Dashboard and designed Tableau dashboards to analyse sales performance, customer trends, and key metrics.

CERTIFICATIONS

- Python for Data Science, AI & Development (Issued by Coursera)
- The Complete SQL Bootcamp: Go from Zero to Hero (Issued by Udemy)
- Tableau Ultimate course: From Zero to Hero (Issued by Udemy)
- Programming Essentials in Python (Issued by OpenEDG Python Institute)
- Python Basics (Issued by Coursera)

EDUCATION

Master of Science in Computer Science, California State University Northridge, Los Angeles, CA

Jan 2023 to Dec 2024

GPA: 3.77 / 4.0

Bachelor of Technology in Electronics and Communication Engineering, GVP college of Engineering, India

Jun 2018 to Jun 2022

GPA: 8.37 / 10.0